. IPL CRICKET ANALYSIS USING DATA SCIENCE

PREPARED BY : VIJAYA GOKUL

1.Introduction:

The Indian Premier League (IPL) is one of the most popular and competitive T20 cricket leagues in the world. Since its inception in 2008, IPL has produced numerous exciting matches and remarkable performances by players and teams. With the availability of large amounts of match data, data science techniques can be used to analyze patterns, identify trends, and generate meaningful insights about the tournament.

This project focuses on analyzing IPL match data using data science and machine learning techniques. By studying historical match data, we aim to understand team performance, player contributions, and match dynamics. The project also includes predictive modeling to estimate match outcomes based on selected features such as teams, toss decisions, and venues.

2. Dataset Description :

The dataset used for this project consists of two main files: **matches.csv** and **deliveries.csv**.

The **matches dataset** contains match-level information such as season, city, teams, toss winner, toss decision, match result, venue, and player of the match.

The **deliveries dataset** contains ball-by-ball details including batsman, bowler, runs scored, extra runs, dismissals, and innings information.

These datasets provide a comprehensive view of IPL matches and allow detailed analysis of both team performance and individual player statistics.

3. Data Preprocessing:

Before performing analysis and predictive modeling, the dataset was cleaned and prepared to ensure accuracy and consistency.

Several preprocessing steps were carried out:

Missing values in columns such as **city, winner, player_of_match, umpire1, and umpire2** were handled by replacing them with appropriate values like "Unknown" or "No Result".

Columns with excessive missing values were removed when necessary to improve data quality.

Categorical variables such as team names, venue, and toss decision were converted into numerical values using **Label Encoding** so that machine learning algorithms could process the data.

These preprocessing steps ensured that the dataset was clean and ready for analysis.

4. Exploratory Data Analysis (EDA)

4.1 Season-wise Match Winners :

This visualization shows the number of matches won by different teams in each IPL season. It helps identify which teams performed consistently across different seasons.

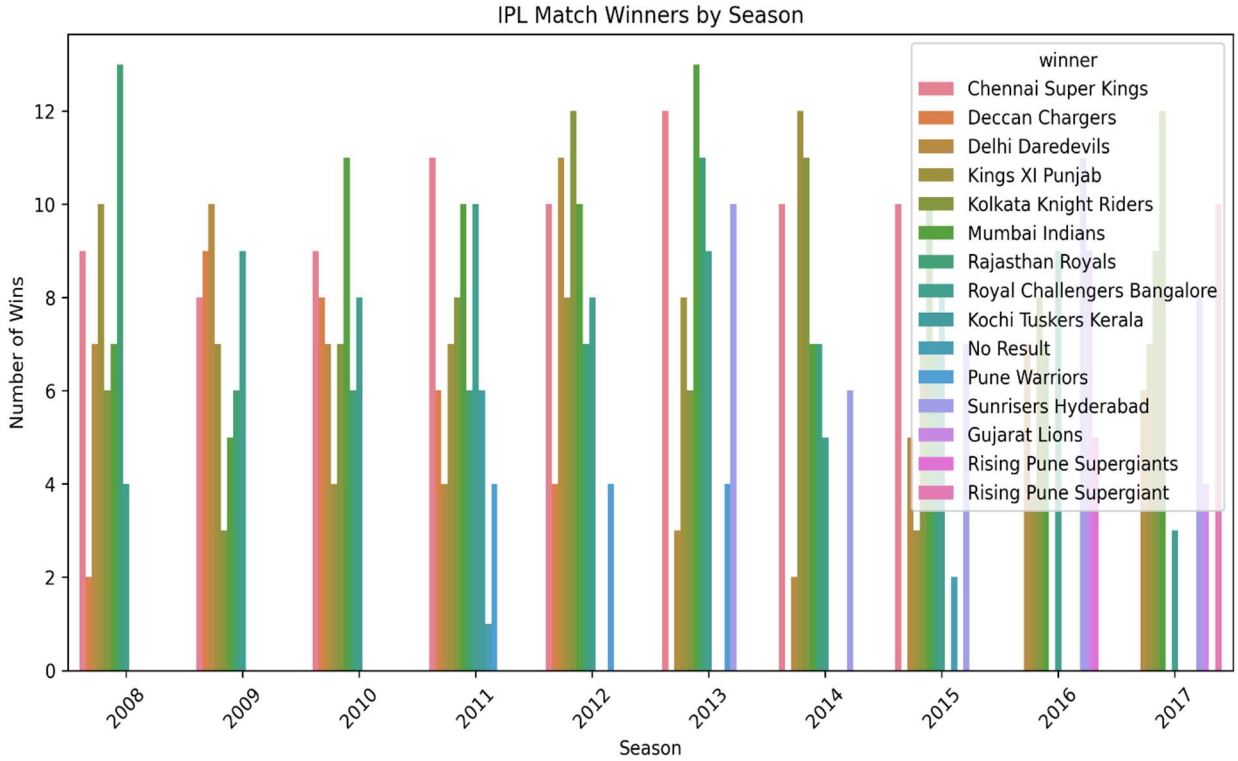


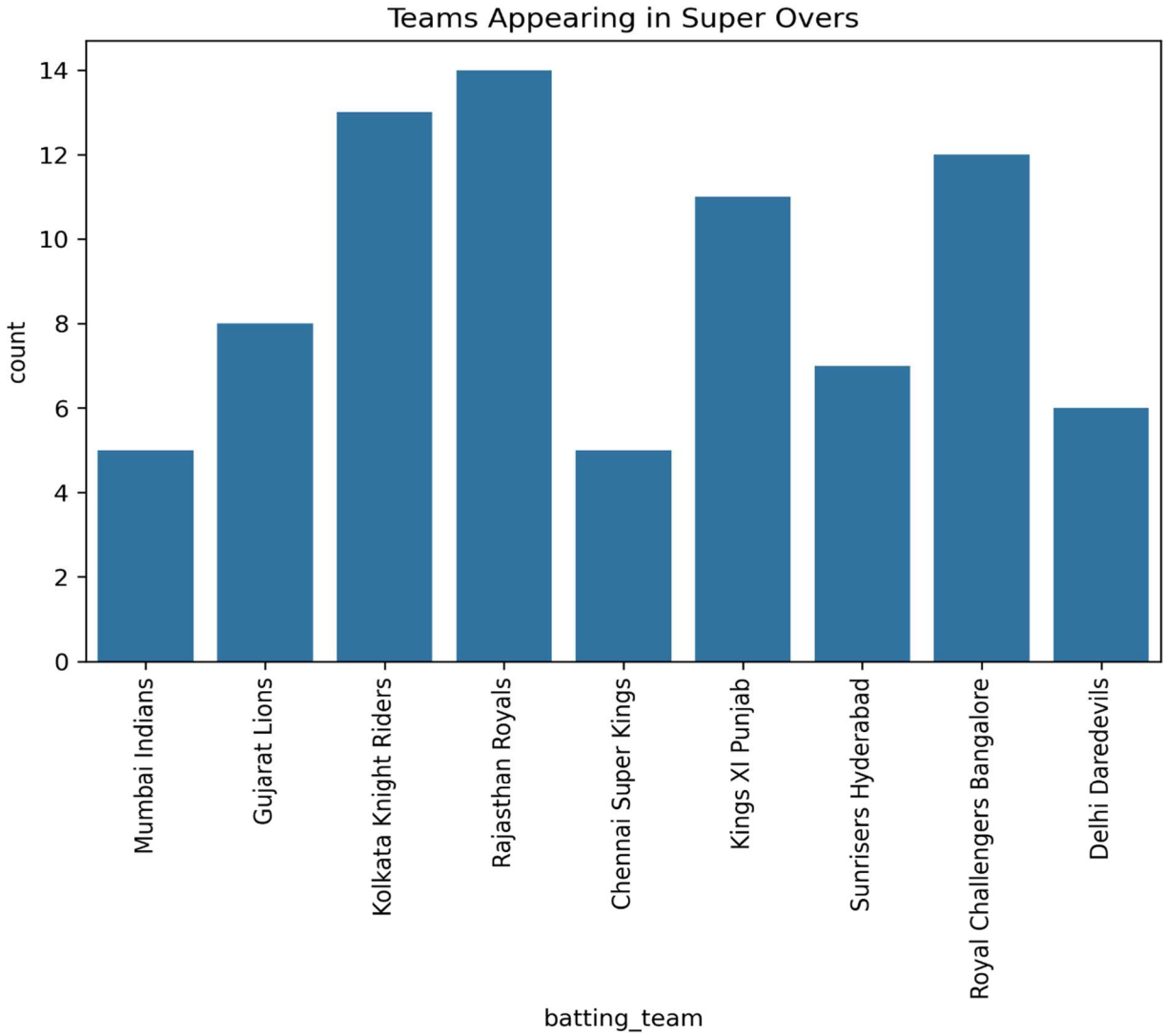
Figure 1 ipl match winners by season

Insight:

The chart highlights the dominance of certain teams in particular seasons and shows variations in team performance across the IPL timeline.

4.2 Super Over Analysis :Figure 2 super over trend analysis

Super overs occur when the match ends in a tie and both teams play an additional over to determine the winner. This analysis identifies matches where super overs were played



. **Figure 2 super over trend analysis**

Insight:

The analysis shows that super overs are rare in IPL matches, indicating that most matches are decided within the regular overs.

4.3 High Scoring Matches (200+ Runs) :

This visualization shows matches where teams scored more than 200 runs in an innings. These matches highlight aggressive batting performances.

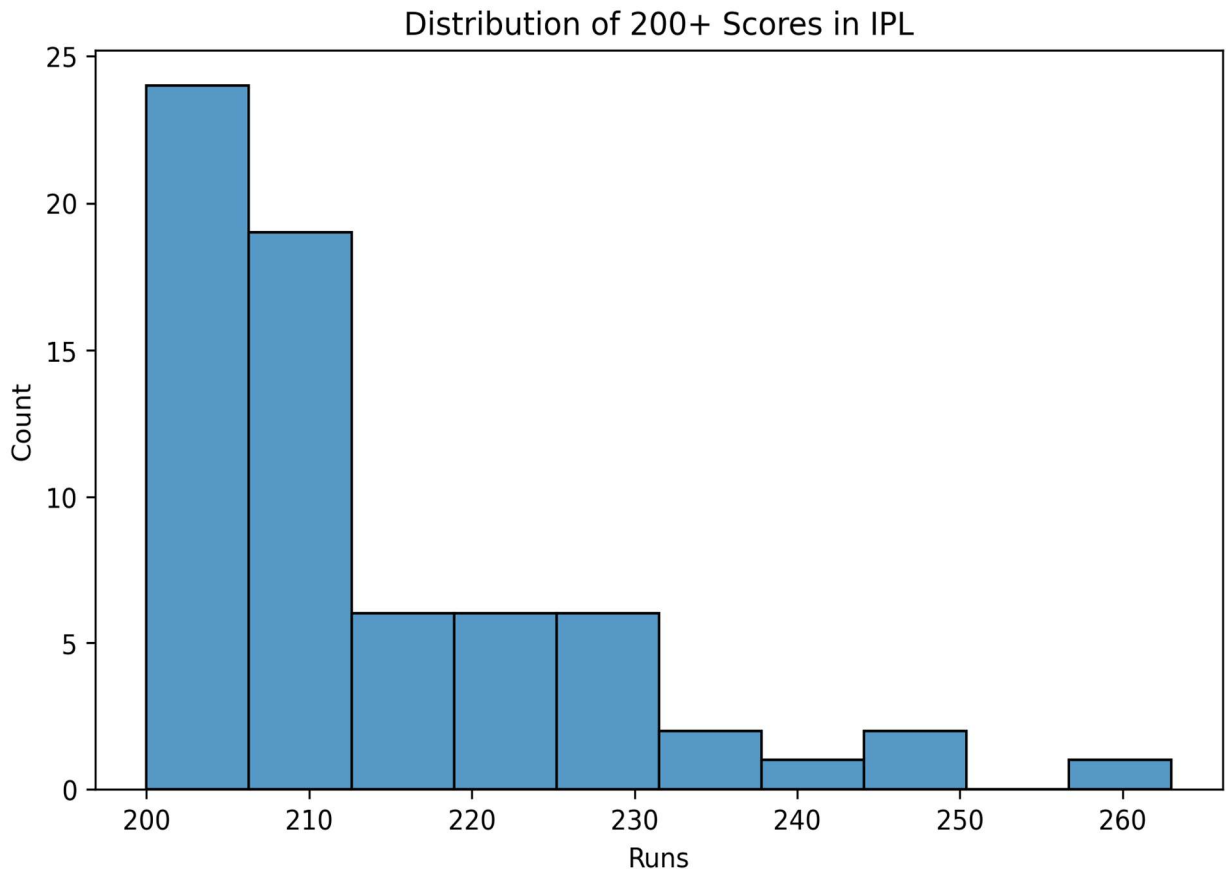


Figure 3 distribution of 200+ scores in ipl

Insight:

The IPL has seen numerous high-scoring innings, reflecting the attacking style of T20 cricket and the impact of strong batting lineups.

4.4Top Bowlers in IPL :

This visualization shows bowlers who have taken the highest number of wickets in IPL matches.

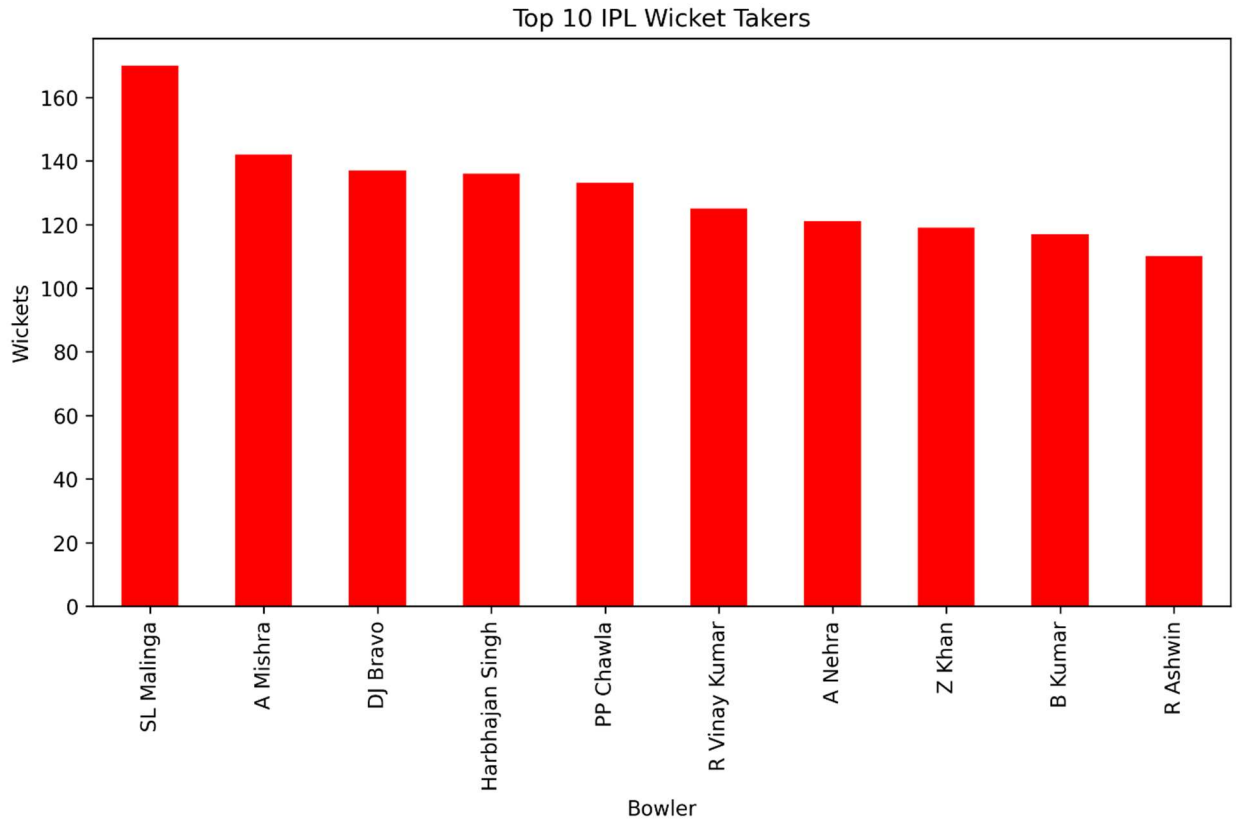


Figure 4 top 10 bowlers

Insight:

The chart reveals bowlers who played key roles in restricting opposition teams and consistently delivered match-winning bowling performances.

z

4.5Top Batsmen in IPL :

This analysis identifies players who have scored the highest number of runs in IPL history

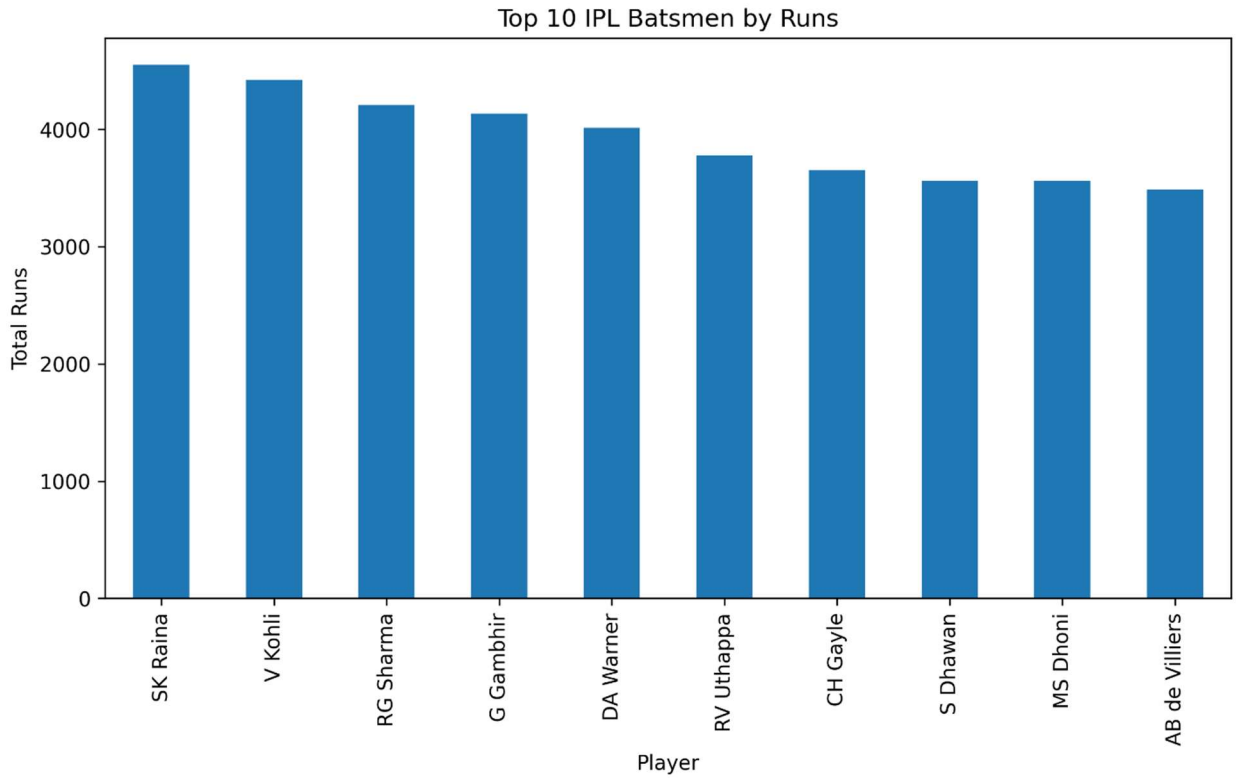


Figure 5 Top ten batsmens

Insight:

The visualization highlights players who consistently contributed to their teams through strong batting performances across multiple seasons.

5. Predictive Modeling :

To predict IPL match outcomes, a **Random Forest Classifier** was implemented. Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their predictions to improve accuracy.

Important features such as **team1, team2, toss winner, toss decision, and venue** were used as inputs to the model.

The dataset was divided into **training and testing sets**, where 80% of the data was used to train the model and 20% was used to evaluate its performance.

The model achieved an accuracy of approximately **55%**, indicating that it could correctly predict match outcomes in more than half of the cases based on historical patterns.

6. Conclusion :

This project demonstrated how data science techniques can be applied to analyze IPL cricket data and uncover meaningful insights. Through exploratory data analysis, several trends related to team performance, player statistics, and match outcomes were identified.

The predictive modeling approach using Random Forest provided a basic framework for predicting match winners based on historical data. Although the model achieved moderate accuracy, it highlights the unpredictable nature of cricket matches where many real-world factors influence results.

Overall, the project shows the potential of data analytics in sports analysis and provides a foundation for further improvements using advanced features and models.